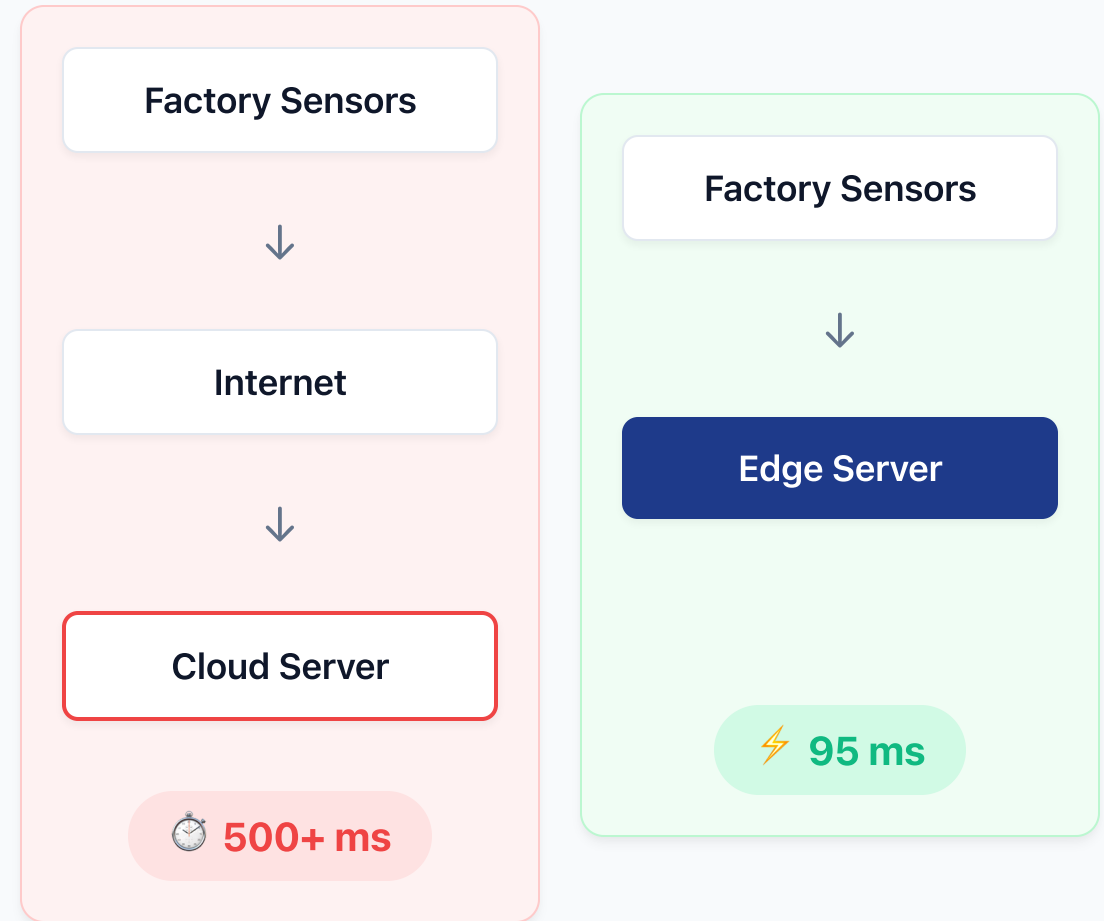


Why Do We Need Edge Computing?

The Problem:

- Billions of IoT devices generate data.
- Industrial systems need millisecond responses.
- Sending everything to the cloud causes delay.
- Delays can damage equipment and reduce safety.

Edge computing brings computation closer to the machines.



The Industrial Problem

Example: Robotic Welding Station

Cloud Processing

Weld defect detected



Data sent to Cloud



Response arrives



Machine Damaged (Too Late)

Edge Processing

Weld defect detected



Local Edge Node



Immediate Shutdown

How Edge Computing Works



Cloud Layer

Long-term storage & Historical Analysis

↑ Only important results go to cloud ↑



Edge Layer

Local processing, immediate actions

↑ Machines generate raw data ↑

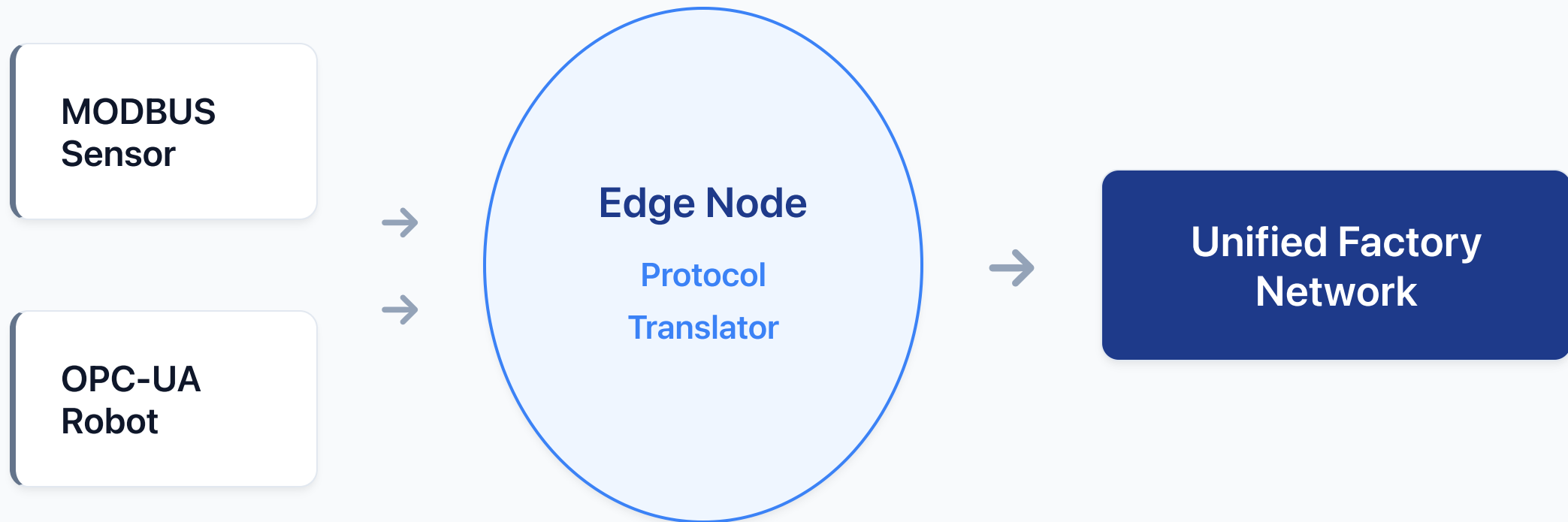


Industrial Devices

Sensors, PLCs, Cameras, Robotic Arms

Edge Node = Factory Translator

Different languages → Edge translates → Everyone communicates



From Experiment to Industrial System

Experiment	Industrial Equivalent
Mobile Phone	Sensor / PLC / Robot
Wi-Fi	Industrial Ethernet / OPC UA / 5G
Ubuntu Laptop	Edge Gateway
Flask Application	Edge Analytics
Latency Measurement	Response Time

Real Industrial Applications



PCB Assembly

Cameras inspect boards instantly. Detects missing components and rejects boards before moving down the line.



CNC Machining

Monitors vibration and motor current. Detects tool wear locally to prevent damage to expensive workpieces.



Robotic Welding

Evaluates voltage and weld times in milliseconds. Stops defective welds immediately.



Pharma Packaging

Verifies fill levels and label positions. Removes faulty or underfilled products directly on the spot.